

Assignment No. 22(optional)

Literature survey of different algorithms used to solve NS Eqns

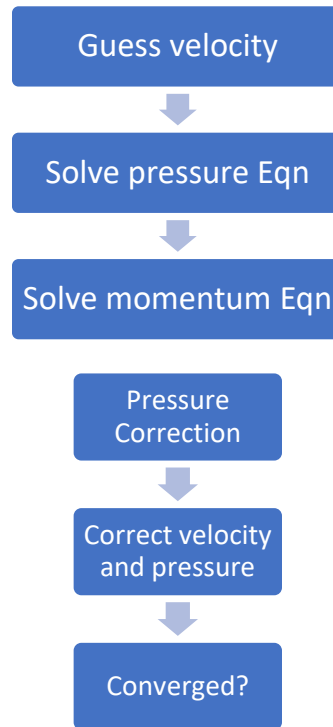
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SIMPLER

Instead of solving for pressure correction(like in SIMPLE), it first solves a pressure Eqn, then solves momentum Eqn, then performs a smaller pressure correction.



Merits: 1. Faster convergence than SIMPLE
2. Better pressure prediction
3. More accurate

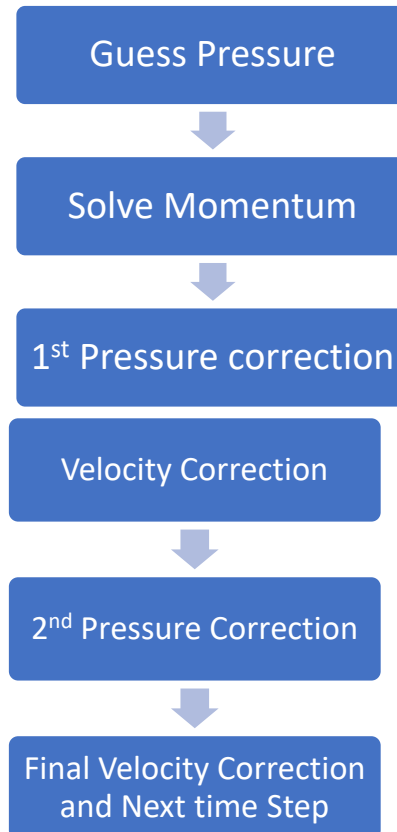
Demerits:

1. More iterations required
2. Slightly more difficult to program than SIMPLE

PISO

Pressure Implicit with Splitting of Operation

Instead of one pressure correction , PISO performs multiple pressure corrections within a single iteration



Merit: 1. Very Fast

2. Excellent for transient flows

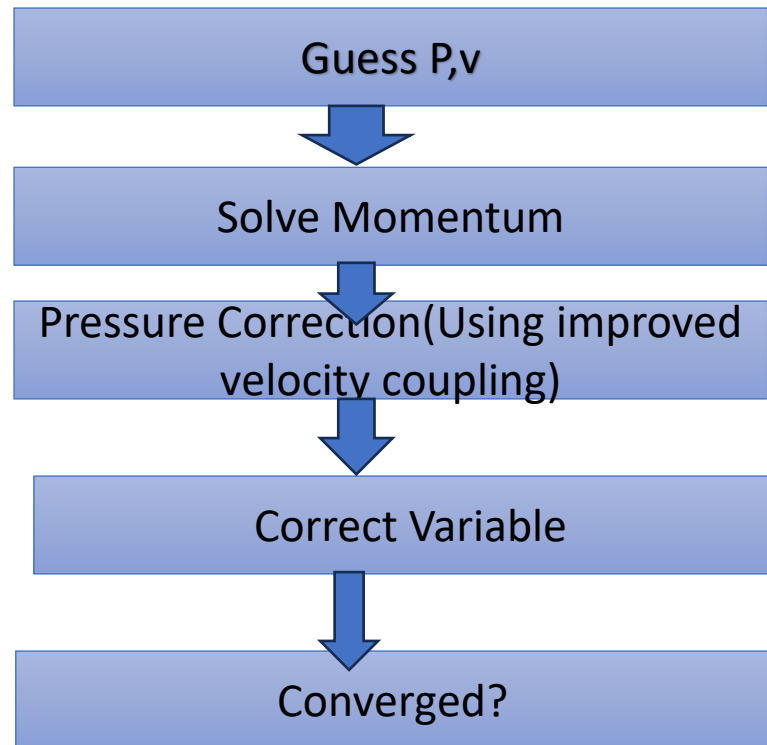
Demerit: 1. More computationally expensive

2. Less efficient for steady state problems

SIMPLEC

SIMPLE Consistent

Modifies the velocity correction equations by neglecting some less important terms, giving stronger velocity coupling. Uses modified velocity correction formula.



Merits: 1. Faster than SIMPLE

2. Fewer iterations

3. Similar implementation

Demerits:

1. Can be less stable for some difficult flows